

Week # 02

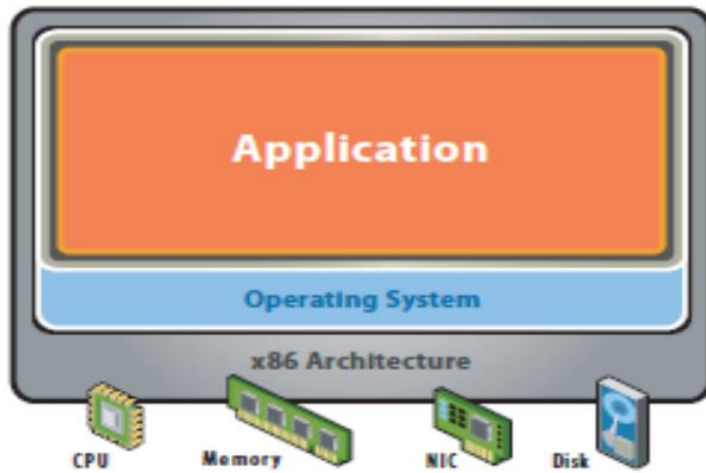
What is Virtualization

- Virtualization abstracts the underlying resources and simplifies their use, isolates users from one another, and supports replication, which, in turn, increases the elasticity of the system.
- Virtualization is a critical aspect of cloud computing, equally important to the providers and consumers of cloud services, and plays an important role in:
 - Performance and reliability because it allows applications to migrate from one platform to another.
 - The development and management of services offered by a provider.
 - Performance isolation.

What is Virtualization

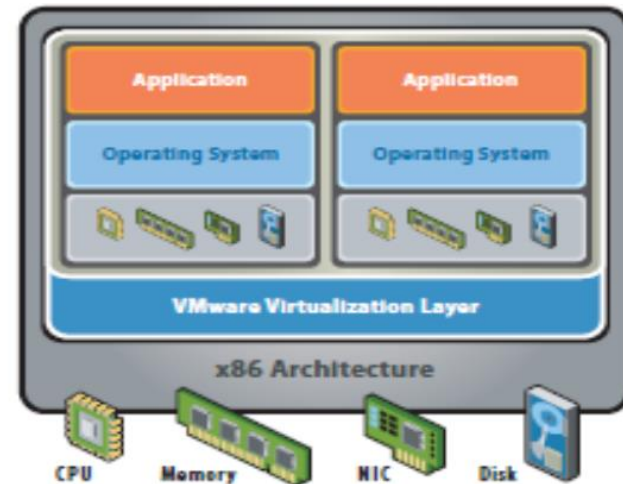
- Virtualization simulates the interface to a physical object by any one of four means:
 - 1. Multiplexing. Create multiple virtual objects from one instance of a physical object. For example, a processor is multiplexed among a number of processes or threads.
 - 2. Aggregation. Create one virtual object from multiple physical objects. For example, a number of physical disks are aggregated into a RAID disk.
 - 3. Emulation. Construct a virtual object from a different type of physical object. For example, a physical disk emulates a random access memory.
 - 4. Multiplexing and emulation. Examples: Virtual memory with paging multiplexes real memory and disk, and a Virtual address emulates a real address; TCP emulates a reliable bit pipe and multiplexes a physical communication channel and a processor.

What is Virtualization



Before Virtualization:

- Single OS image per machine
- Software and hardware tightly coupled
- Running multiple applications on same machine often creates conflict
- Underutilized resources
- Inflexible and costly infrastructure



After Virtualization:

- Hardware-independence of operating system and applications
- Virtual machines can be provisioned to any system
- Can manage OS and application as a single unit by encapsulating them into virtual machines

Virtualization Concept



Virtual Resources

- ☒ Substitutes for real resources: **same interfaces/functions, different attributes.**
- ☒ Often of part of the underlying resource, but may span multiple resources.

Virtualization – a substitution process

- ☒ Creates virtual resources from real resources.
- ☒ Primarily accomplished with software and/or firmware.

Resources

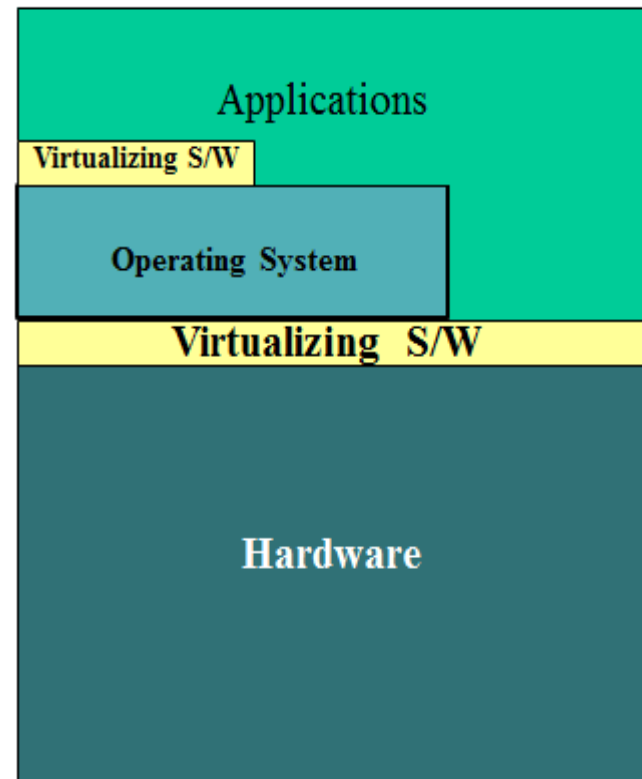
- ☒ Components with **architected interfaces/functions.**
- ☒ Usually physical. May be centralized or distributed.
- ☒ Examples: memory, disk drives, networks, servers.



- ☒ Separates presentation of resources to users from actual resources
- ☒ Aggregates pools of resources for allocation to users as virtual resources

Virtualization how

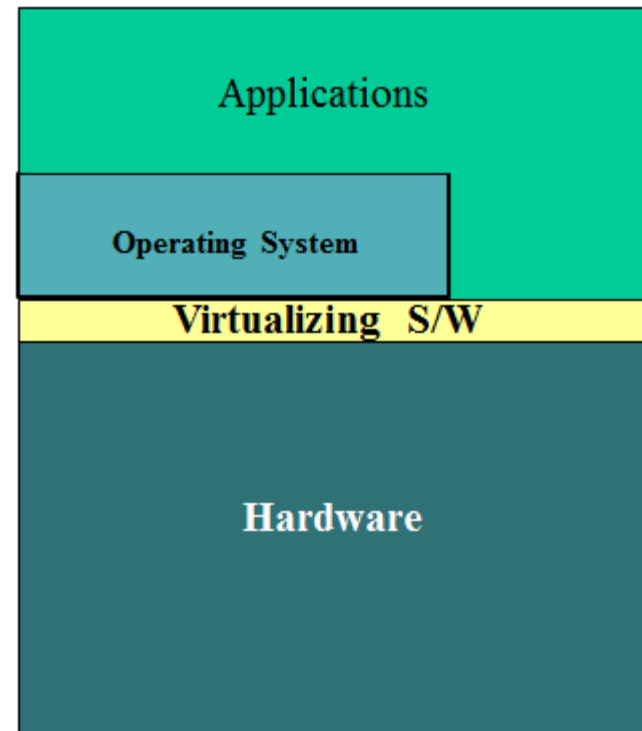
- Virtualization... How?
- By adding a layer between execution stack layers.
- Types of Virtualization



Virtualization how

• Hardware-level virtualization

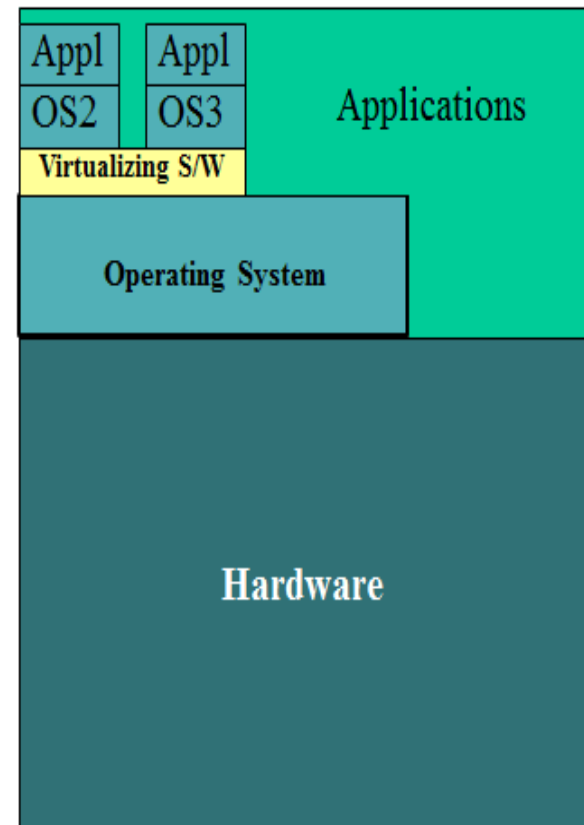
- VMM* is placed between the hardware and the OS.
- Could provide a different ISA.
- e.g. Emulators.
- Tasks:
 - Maps virtual resources to real ones.
 - Translate virtual instructions to real one
 - * VMM = Virtual Machine Monitor



Virtualization how

- **System-level virtualization**

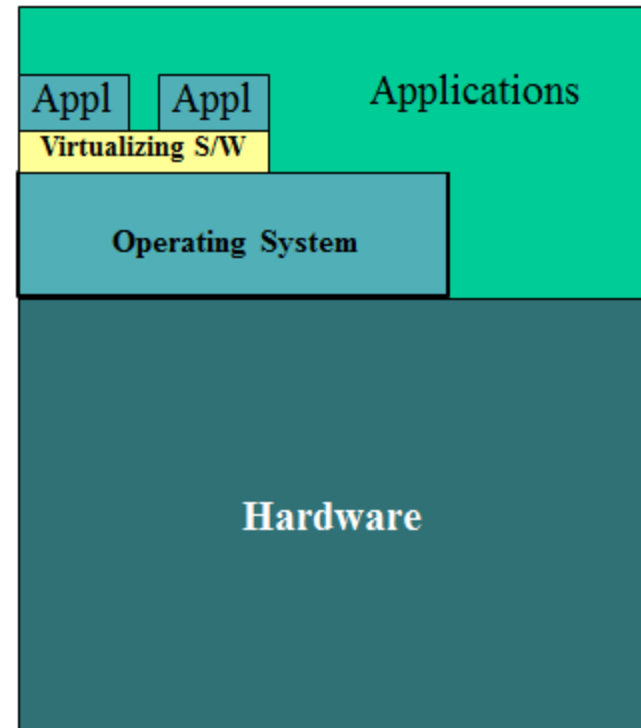
- placed between the OS and other OS e.g. VMware Wks and UML
- Enable several OS to run on single hardware.
- Terminology
 - Host OS
 - Guest OS
- Guest OS and VM run in Application privilege ring.



Virtualization how

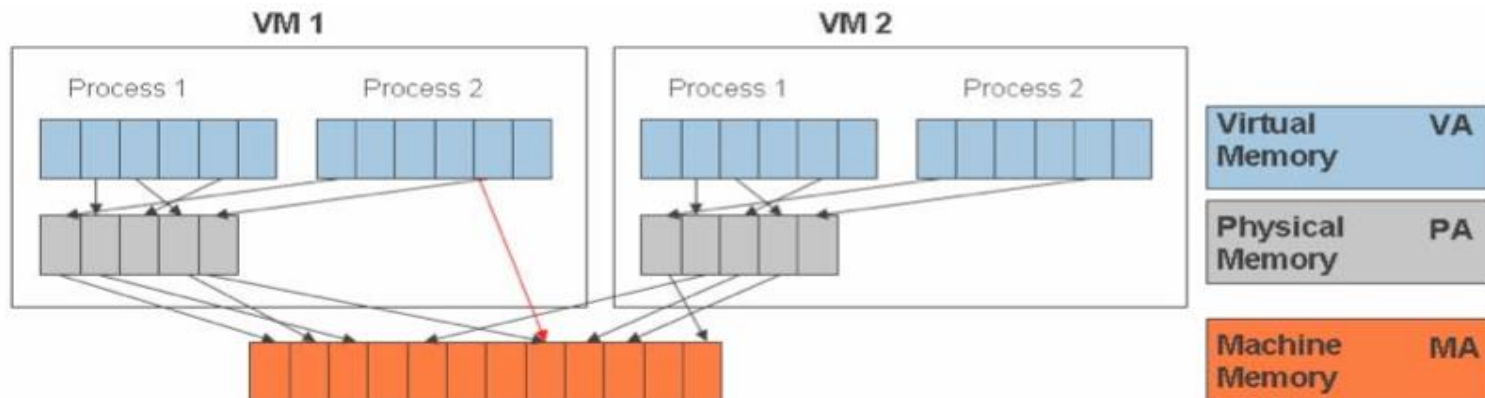
•Application-level virtualization

- VM is placed between the OS and the applications.
- e.g. Java Virtual Machine (JVM)
Provide same interface to all
- Application, irrespective of OS.
- Provide Application Sand-boxing.
- Tasks:
 - Translate Application byte code to
 - OS-specific executable.



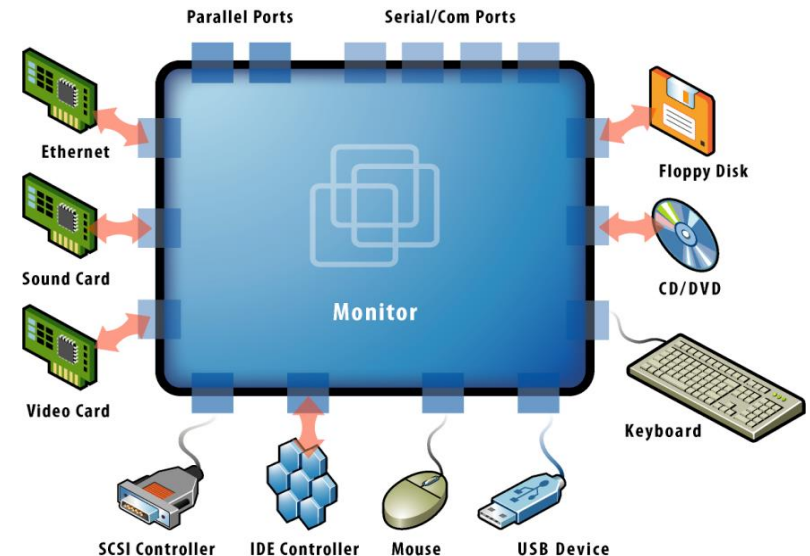
Memory virtualization

- Beyond CPU virtualization, This involves sharing the physical system memory and dynamically allocating it to virtual machines.
- The operating system keeps mappings of virtual page numbers to physical page numbers stored in page tables.
- All modern x86 CPUs include a memory management unit (MMU) and a translation look-aside buffer (TLB) to optimize virtual memory performance.



Device and I/O Virtualization

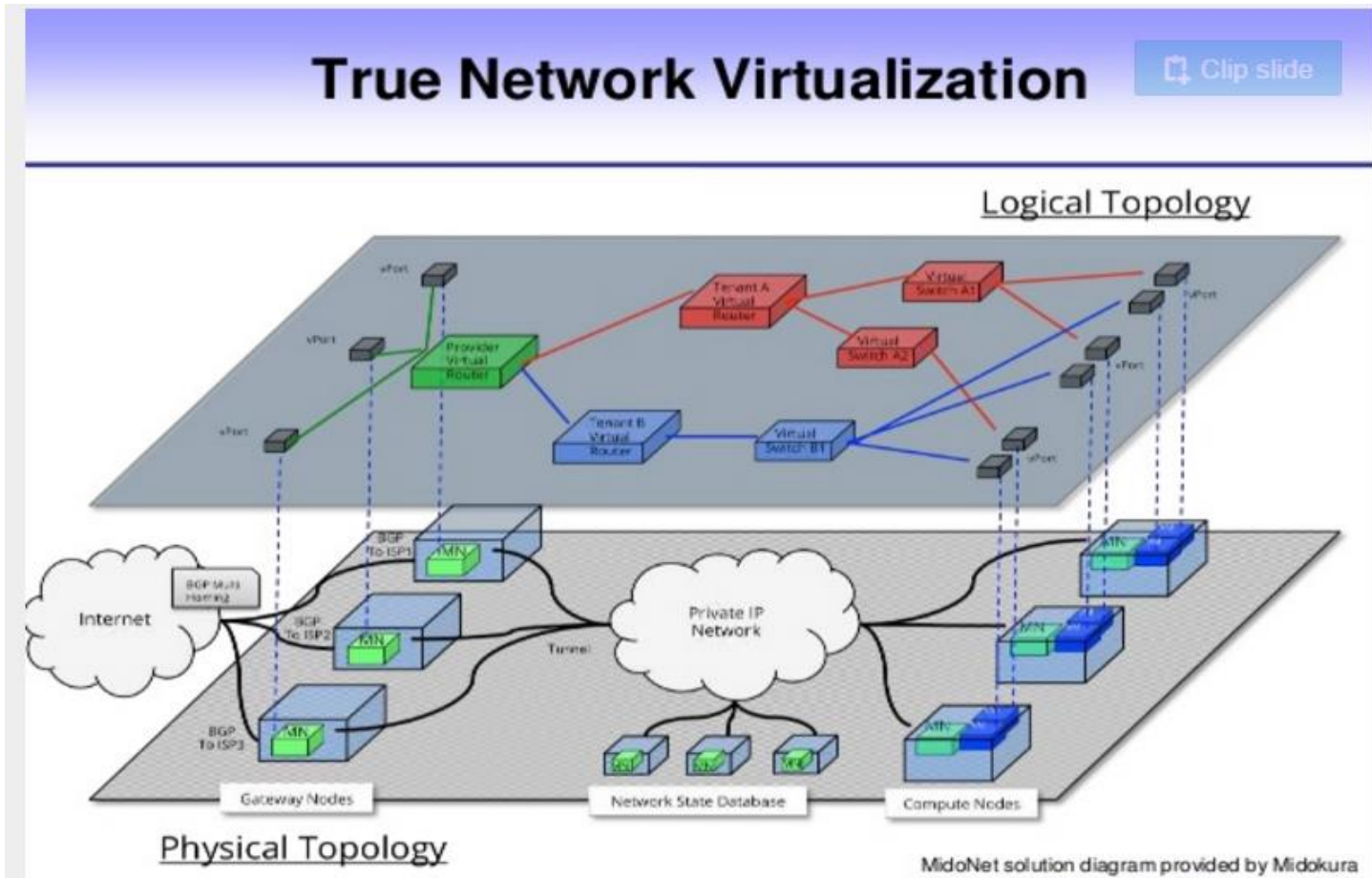
- The final component required beyond CPU and memory virtualization is device and I/O virtualization.
- This involves managing the routing of I/O requests between virtual devices and the shared physical hardware
- The hypervisor virtualizes the physical hardware and presents each virtual machine with a standardized set of virtual devices as seen in Figure



Network Functions Virtualization

- Network functions virtualization (NFV) (also known as virtual network function (VNF)) offers a new way to design, deploy and manage networking services.
- NFV decouples the network functions, such as network address translation (NAT), firewalling, intrusion detection, domain name service (DNS), and caching, to name a few, from proprietary hardware appliances so they can run in software.
- It's designed to consolidate and deliver the networking components needed to support a fully virtualized infrastructure – including virtual servers, storage, and even other networks.
- It utilizes standard IT virtualization technologies that run on high-volume service, switch and storage hardware to virtualize network functions.
- It is applicable to any data plane processing or control plane function in both wired and wireless network infrastructures

Network Functions Virtualization



Virtual machine monitors/ Hypervisors

- A virtual machine monitor (VMM), also called a hypervisor, is the software that securely partitions the resources of a computer system into one or more virtual machines.
- A guest operating system is an operating system that runs under the control of a VMM rather than directly on the hardware.
- The VMM runs in kernel mode, whereas a guest OS runs in user mode.
- Sometimes the hardware supports a third mode of execution for the guest OS.
- VMMs allow several operating systems to run concurrently on a single hardware platform; at the same time, VMMs enforce isolation among these systems, thus enhancing security.

Virtual machine monitors/ Hypervisors

(VMM). When VMMs are combined with the modules that provide OS-level services, such as the scheduling of VMs in the CPU, they are called the hypervisor. These hypervisor baseline features or functions are:

- HY-BF1: VM Process Isolation – Provides scheduling of VMs for execution, Management of the application processes running in VMs such as CPU and Memory Management, and context switching between various processor states during the running of applications in VMs. In order to ensure VM process isolation, memory access from Direct Memory Access (DMA) capable devices needs to be under hypervisor control as well (e.g., via Input Output Memory Management Unit (IOMMU)). However, this function is considered under HY-BF2 since it pertains to devices mediation.
- HY-BF2: Devices Mediation and Access Control – Makes devices available to VMs (e.g., via emulation, para-virtualization, passthrough or self-virtualizing hardware devices) and controlling which VMs are allowed to access which devices (e.g., Network Interface Card (NIC), storage device such as IDE drive, etc.).
- HY-BF3: Direct Execution of commands from Guest VMs – Certain commands from Guest OSs are executed directly by the hypervisor instead of being triggered through interrupts and context switching. This function applies to hypervisors that have implemented para-virtualization instead of full virtualization
- HY-BF4: VM Lifecycle Management – All functions including creation and management of VM images, control of VM states (Start, Pause, Stop), VM migration, making snapshots, VM monitoring, and policy enforcement
- HY-BF5: Management of hypervisor platform – Defining artifacts and setting values for various configuration parameters in hypervisor software modules including those for configuration of a Virtual Network inside the hypervisor and updates and patching to those modules.

Virtual machine monitors/ Hypervisors

Table 1: Hypervisor Platform Baseline functions

Baseline function	Component (Software Module)	Location
VM Process Isolation (HY-BF1)	Hypervisor Kernel	Either an OS kernel (along with a kernel module) itself or a component installed on a full-fledged OS (Host OS)
Devices Mediation and Access Control (HY-BF2)	Device emulator or Device driver	Either in a dedicated VM (called Device-driver VM) or in the hypervisor kernel itself
Direct Execution of commands from Guest VMs (HY-BF3)	Hypervisor Kernel	Pertains to only para-virtualized hypervisors and handled by hypercall interfaces in that type of hypervisor
VM Lifecycle Management (HY-BF4)	A management daemon	Installed on top of hypervisor kernel but runs in unprivileged mode
Management of hypervisor platform (HY-BF5)	A set of tools with CLI (command line interface) or a GUI (Graphical User Interface)	A console or shell running on top of hypervisor kernel

Virtual machine monitors/ Hypervisors

- A VMM controls how the guest operating system uses the hardware resources. The events occurring in one VM do not affect any other VM running under the same VMM.
- At the same time, the VMM enables:
 - Multiple services to share the same platform.
 - The movement of a server from one platform to another, the so-called live migration.
 - System modification while maintaining backward compatibility with the original system.
- When a guest OS attempts to execute a privileged instruction, the VMM traps the operation and enforces the correctness and safety of the operation.
- The VMM guarantees the isolation of the individual VMs, and thus ensures security and encapsulation, a major concern in cloud computing.

Virtual machine monitors/ Hypervisors

- At the same time, the VMM monitors system performance and takes corrective action to avoid performance degradation; for example, the VMM may swap out a VM(copies all pages of that VM from real memory to disk and makes the real memory frames available for paging by other VMs) to avoid thrashing.
- A VMM virtualizes the CPU and memory. For example, the VMM traps interrupts and dispatches them to the individual guest operating systems.
- If a guest OS disables interrupts, the VMM buffers such interrupts until the guest OS enables them.
- The VMM maintains a shadow page table for each guest OS and replicates any modification made by the guest OS in its own shadow page table.
- This shadow page table points to the actual page frame and is used by the hardware component called the memory management unit (MMU) for dynamic address translation.

Type 1 hypervisors

- **Type 1 hypervisor** is installed directly on bare-metal hardware, it doesn't require an additional OS, it is the OS, even it is a light or minimal OS
- Examples:
 - HyperOne, Xen, PikeOS, OKL4, Vmware ESX
- **Advantages:**
 - System is thin, the hypervisor has direct access to the HW
 - higher density hardware.
- **Disadvantages:**
 - HW should support virtualization technology,
 - costlier and Really bad console interface.

Type 1 hypervisors

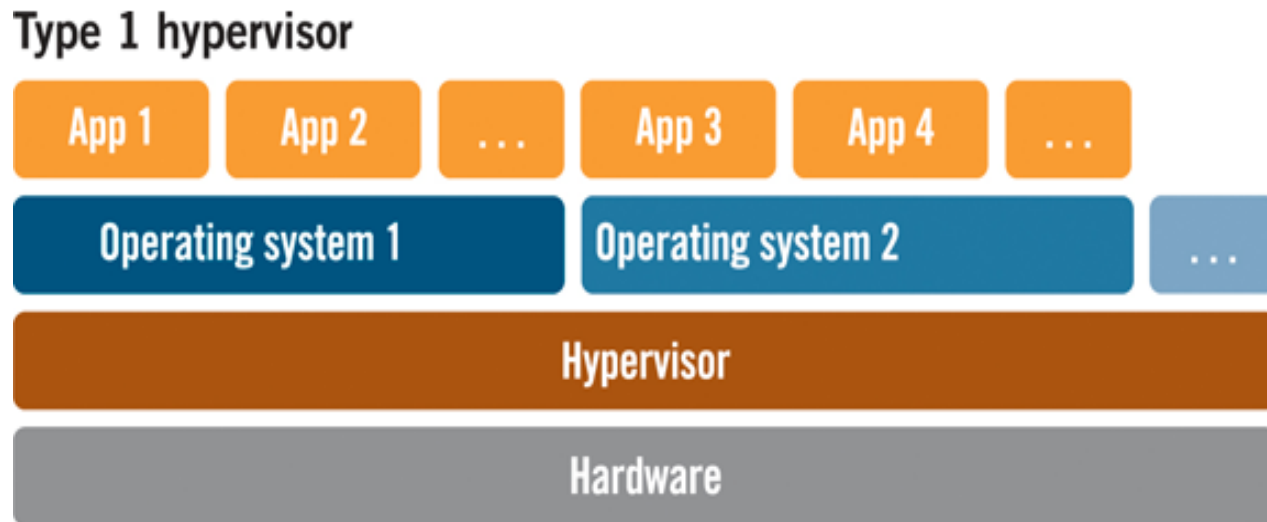


Figure 2. A Type 1 or bare-metal hypervisor sits directly on the host hardware.

Type2 Hypervisors

- **Type 2 hypervisor**

- Type 2 is more of an application installed on an operating system and not directly on the bare-metal.

- EX. VirtualBox and Vmware Workstation

- **Advantages:**

- Run on a greater array of HW because the underlying Host OS is controlling HW access,
- Easy user interface, Data can be secured on the desktop

- **Disadvantages:**

- Decreased security, Loss of Centralized Management, Lower VM Density,
- Cannot support as many VMs as the first type.

Type2 Hypervisors

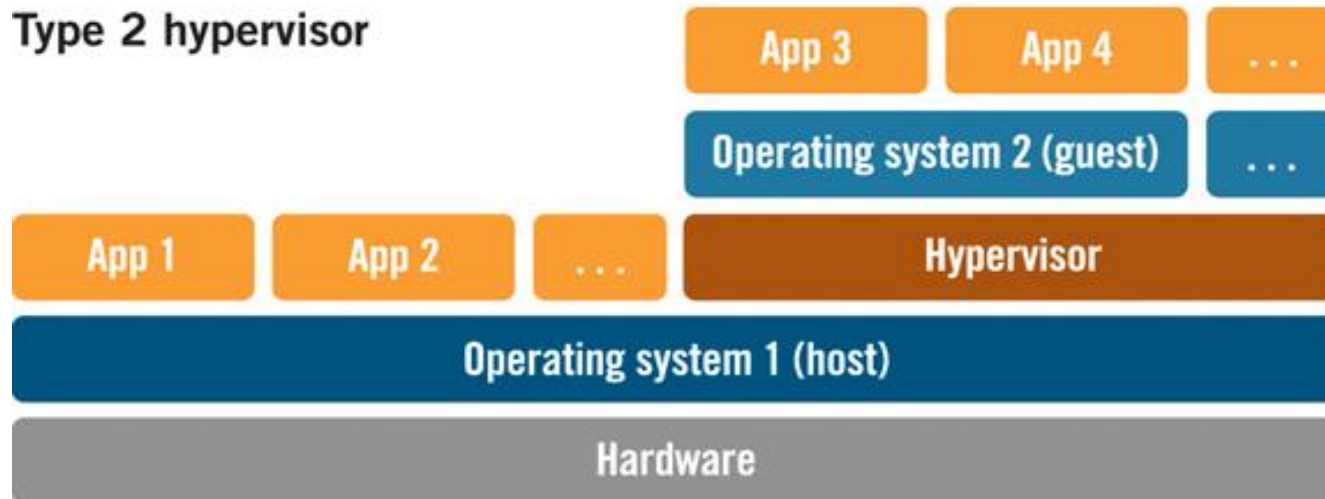


Figure 1. A Type 2 hypervisor runs as an application on a host operating system.

Why Need Virtualization?

- Virtualization is being used by a growing number of organizations to reduce power consumption and air conditioning needs and trim the building space and land requirements that have always been associated with server farm growth.
- Virtualization also provides high availability for critical applications, and streamlines application deployment and migrations.
- Virtualization can simplify IT operations and allow IT organizations to respond faster to changing business demands.

Challenges of virtualization

- Performance degradation -
 - As it interposes and abstraction layer between guest & host.
- Inefficiency and degraded user experience
 - Some of specific features of the host is unexposed.
- Security holes and new threats
 - Case 1 – emulating a host in a completely transparent manner.
 - Case 2 - H/w virtualization , malicious programs can preload themselves before the OS and act as a thin VMM.

Major Vendors of Hypervisor



Microsoft
Hyper-V

CITRIX XenServer

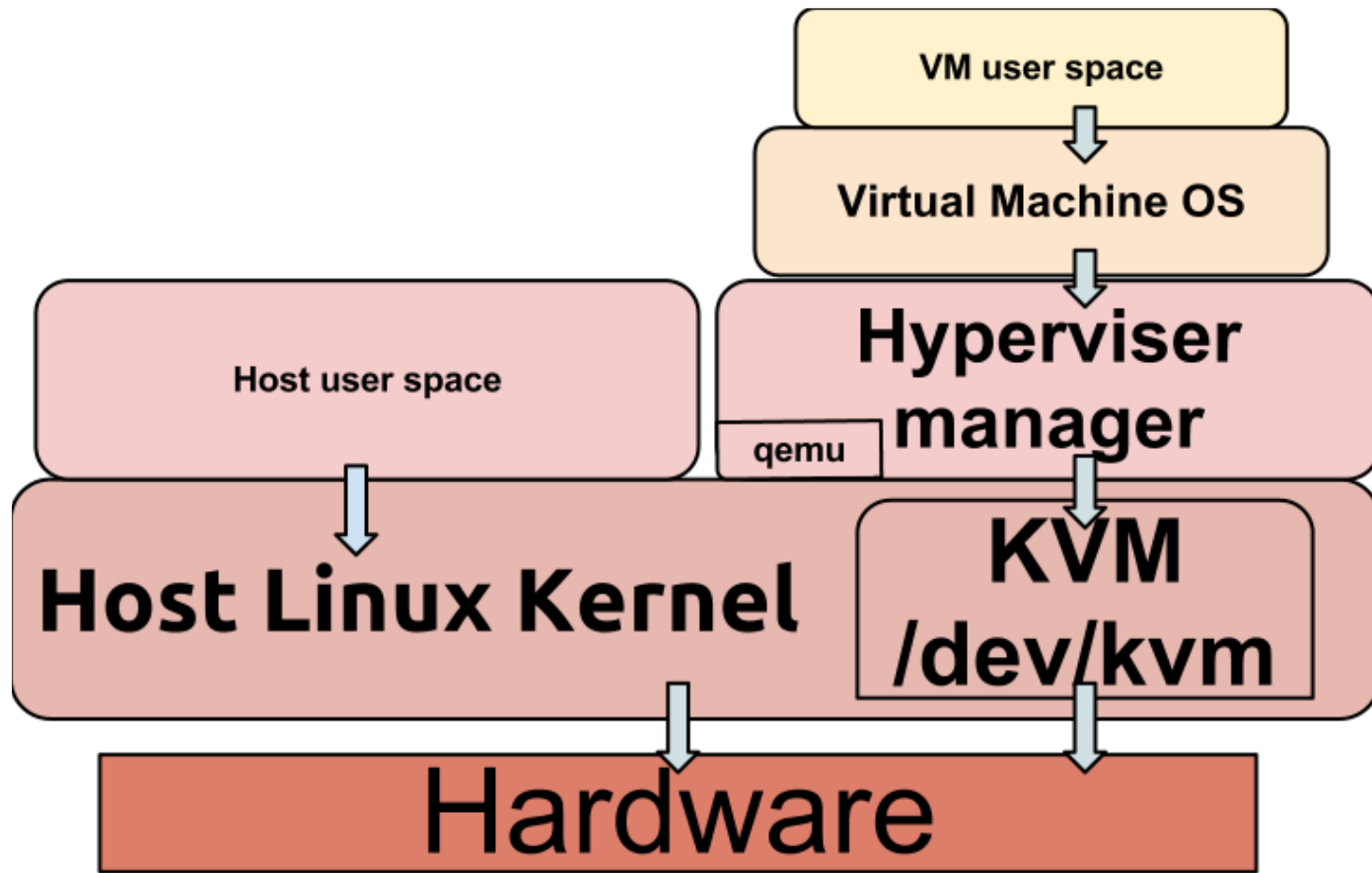
VMware vSphere

KVM hypervisor



- KVM (for Kernel-based Virtual Machine) is a full virtualization solution for Linux on x86 hardware containing virtualization extensions (Intel VT or AMD-V).
- It consists of a loadable kernel module, `kvm.ko`, that provides the core virtualization infrastructure and a processor specific module, `kvm-intel.ko` or `kvm-amd.ko`.
- Using KVM, one can run multiple virtual machines running unmodified Linux or Windows images
- Each virtual machine has private virtualized hardware: a network card, disk, graphics adapter, etc.
- KVM is open source software. The kernel component of KVM is included in mainline Linux, as of 2.6.20.
- The userspace component of KVM is included in mainline QEMU, as of 1.3.

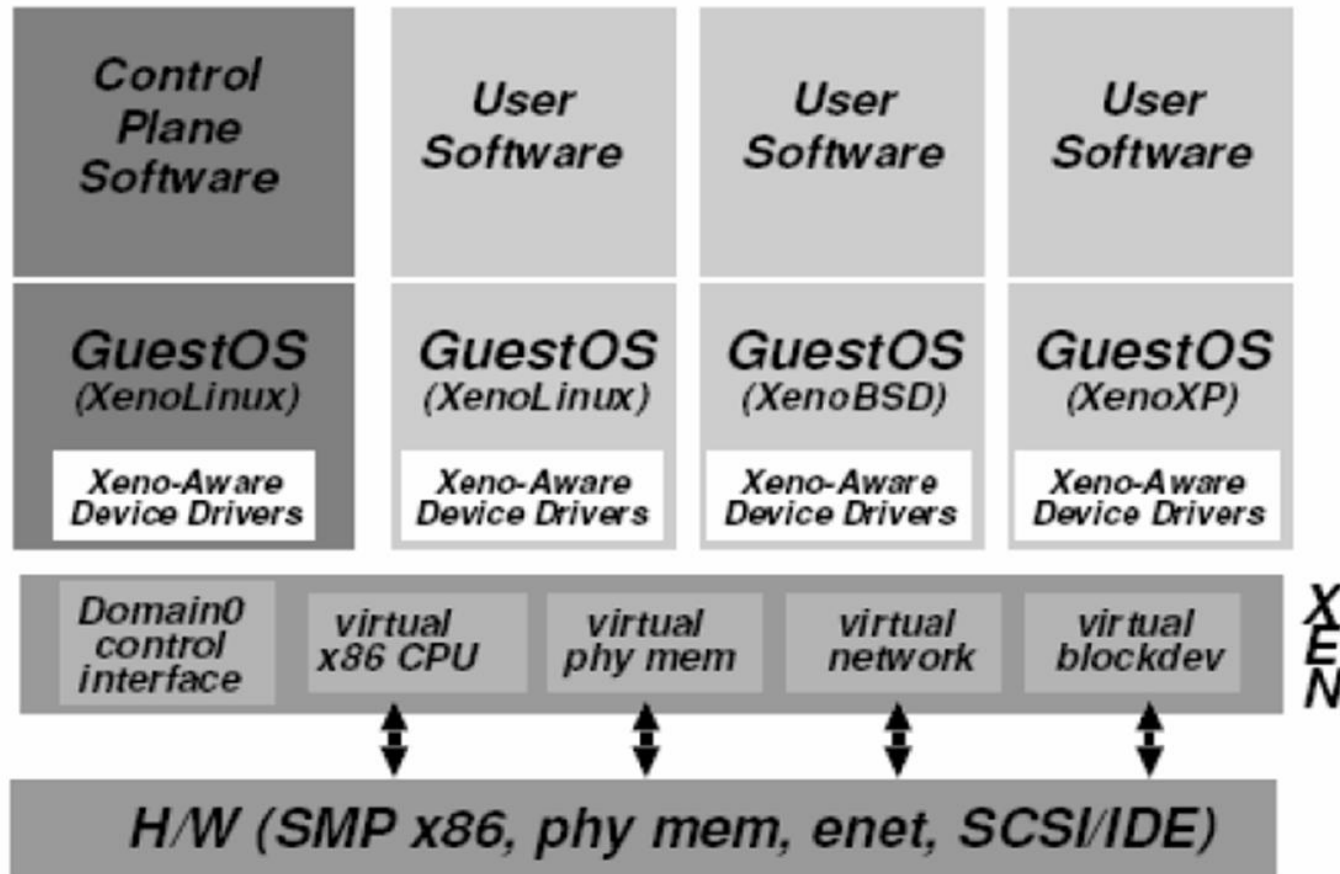
KVM hypervisor



Citrix xenServer

- Citrix XenServer is a server virtualization platform based on the Xen hypervisor that allows IT administrators to host, deploy and manage virtual machines.
- Prior to XenServer 6.2, IT administrators could download only the base hypervisor for free and deploy as many VMs as they wanted, but the hypervisor lacked many key features such as storage integration and strong load balancing.
- Version 6.5 extended some of the capabilities from the Enterprise and Platinum editions to the basic edition, but that version is no longer completely free.

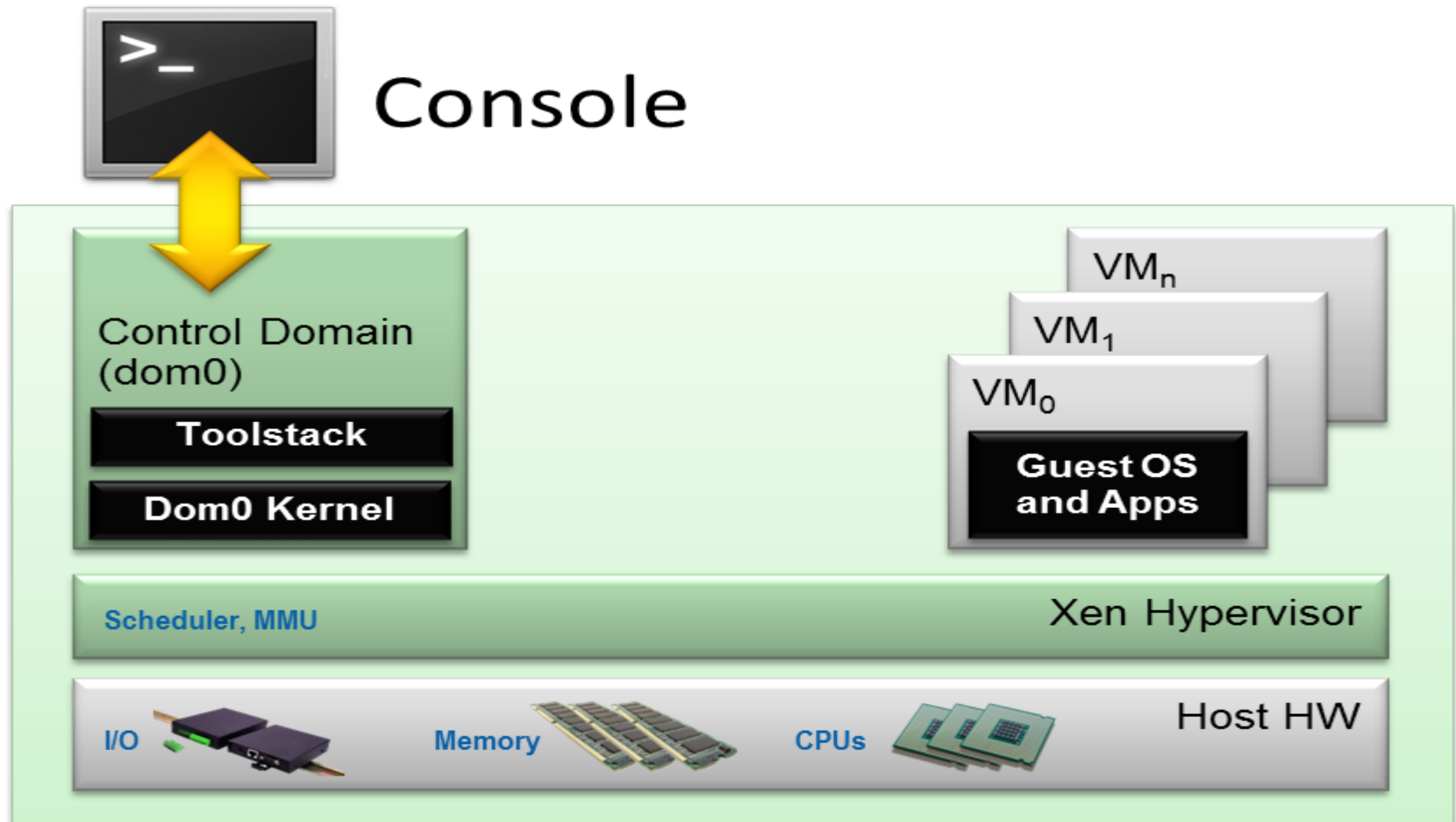
citrix Xen hypervisor



XEN Hypervisor architecture

- Guest virtual machines running on a Xen Project Hypervisor are known as “domains”
- Special domain known as dom0 is responsible for controlling the hypervisor and starting other guest operating systems
- These other guest operating systems are called domUs, this is because these domains are “unprivileged” in the sense they cannot control the hypervisor or start/stop other domains.

XEN Hypervisor architecture



VMware

- VMware is a global leader in cloud infrastructure and business mobility.
- VMware accelerates customers' digital transformation journey by enabling enterprises to master a software-defined approach to business and IT.
- With VMware solutions, organizations are creating exceptional experiences by mobilizing everything, responding faster to opportunities with modern data and apps hosted across hybrid clouds, and safeguarding customer trust with a defense-in-depth approach to cyber-security.
- We will be focusing on
 - **Vmware Workstation**
 - **Vmware ESXi**

VMWare ESXi

- VMware ESXi is the industry-leading, purpose-built bare-metal hypervisor.
- ESXi installs directly onto your physical server enabling it to be partitioned into multiple logical servers referred to as virtual machines

VMWare ESXi Features

- **Improved Reliability and Security**

- The ESXi bare-metal hypervisor's management functionality is built into the VMkernel, reducing the footprint to 150 MB. This gives it a very small attack surface for malware and over-the-network threats, improving reliability and security.

- **Streamlined Deployment and Configuration**

- With few configuration options and simple deployment and configuration, the ESXi architecture makes it easy to maintain a consistent virtual infrastructure.

- **Reduced Management Overhead**

- vSphere ESXi uses an agentless approach to hardware monitoring and system management with an API-based partner integration model. Management tasks are on remote command lines with the vSphere Command Line Interface (vCLI) and Power CLI, which uses Windows PowerShell cmdlets and scripts for automated management.

VMWare ESXi Features

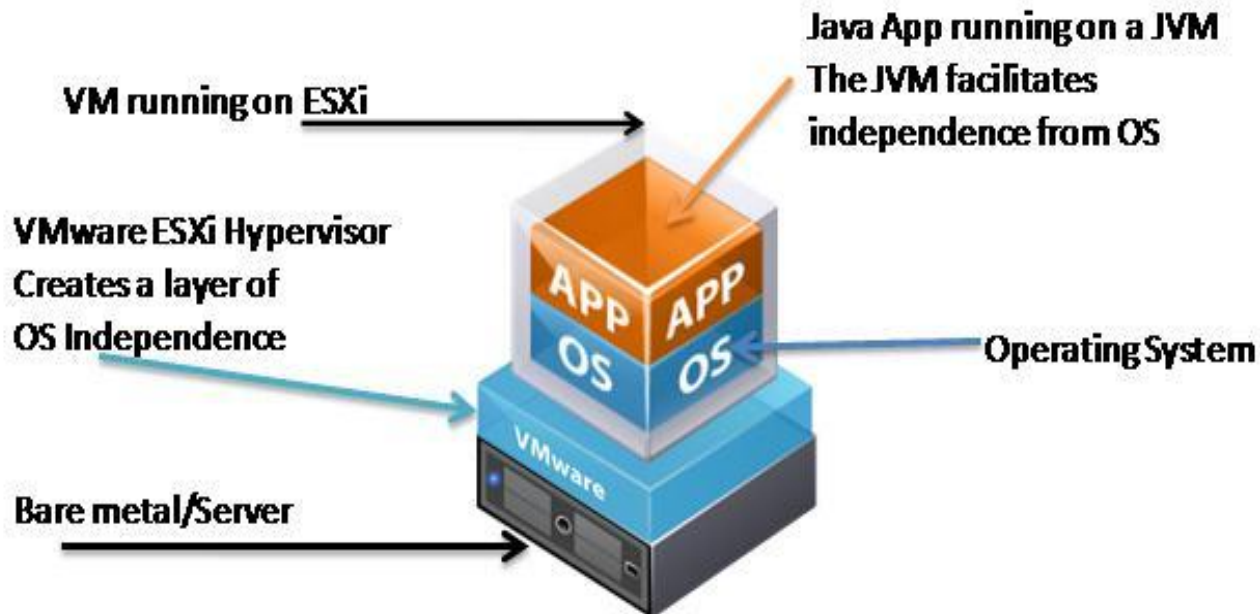
- **Improved Security**

- Improved role-based access and control eliminates the dependency on a shared root account. Users and groups can be assigned full administrative privileges. There's no need to share access or a common super user account in order to perform administrative tasks.

- **Extensive Logging and Auditing**

- vSphere ESXi logs all user activity from both the Shell and the Direct Console User Interface under the user's account. This logging ensures user accountability and makes it easy to audit user activity.

Vmware Esxi Block Diagram



Vmware esxi

- VMWare ESXi
- Type-1 Hypervisor

```
VMware ESXi 5.0.0 (VMKernel Release Build 469512)  
VMware, Inc. VMware Virtual Platform  
2 x Intel(R) Core(TM) i7 CPU 920 @ 2.67GHz  
2 GiB Memory
```

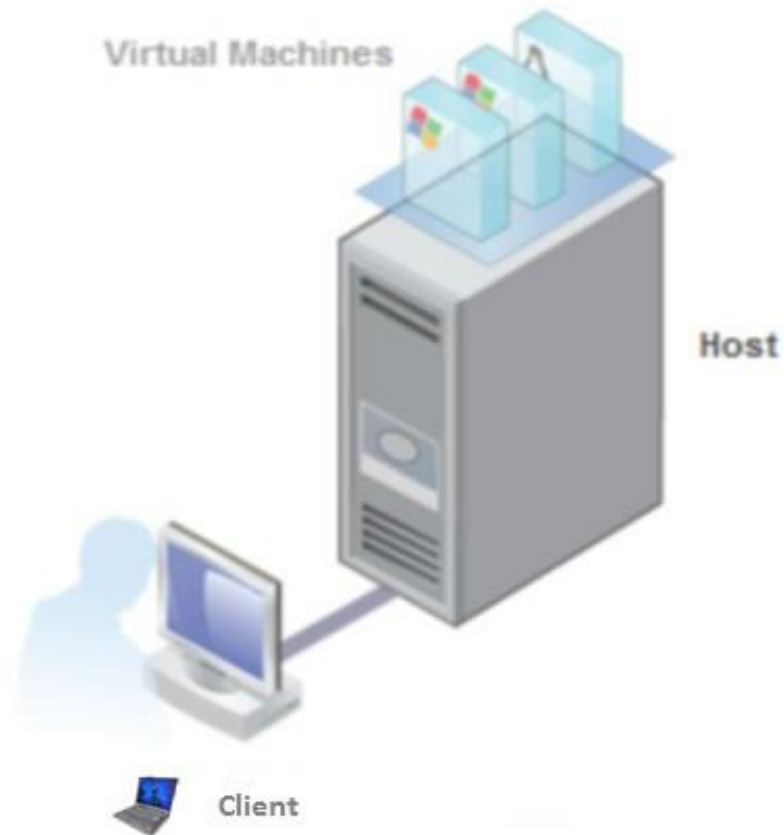
```
Download tools to manage this host from:  
http://192.168.1.38/ (DHCP)
```

<F2> Customize System/View Logs

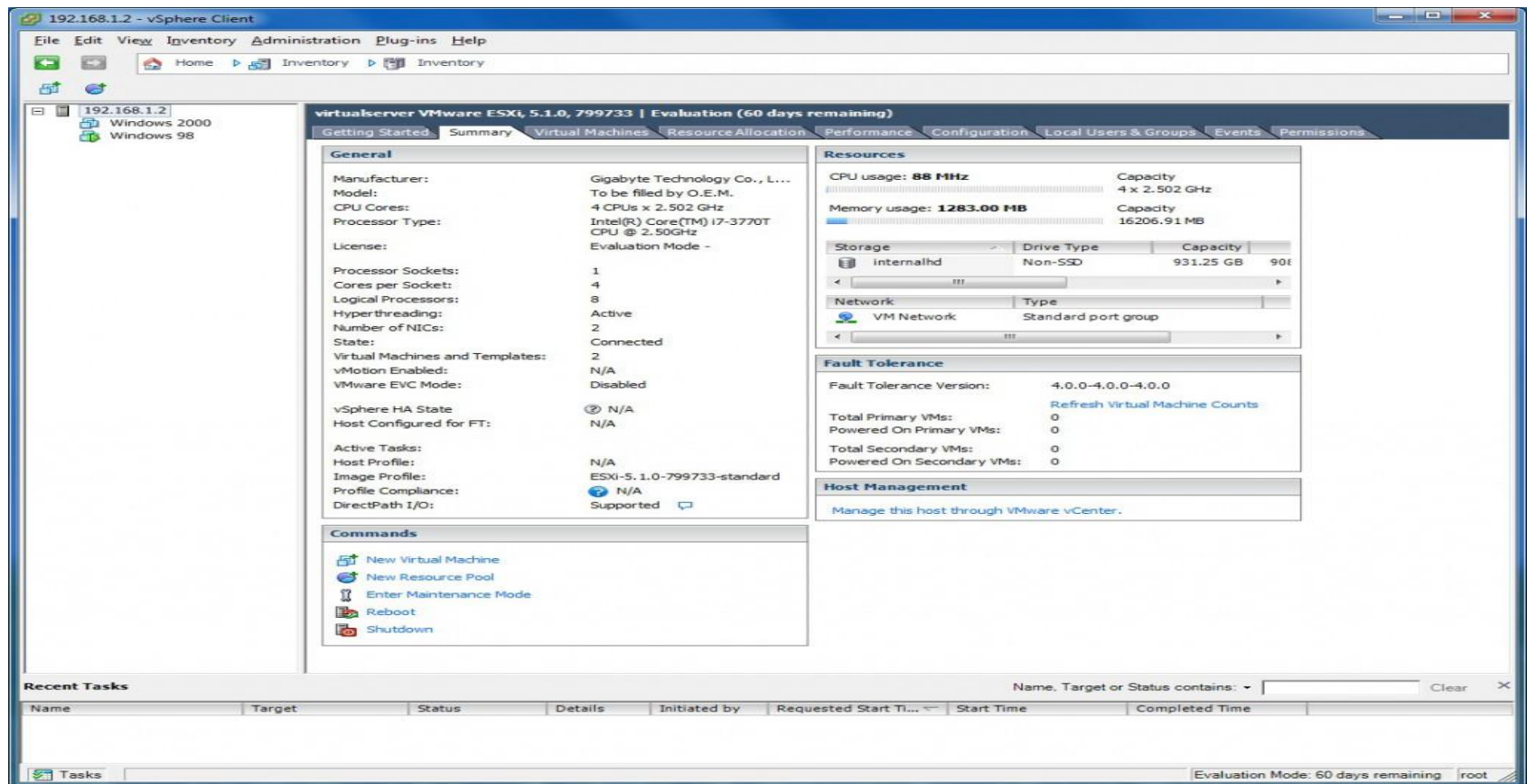
<F12> Shut Down/Restart

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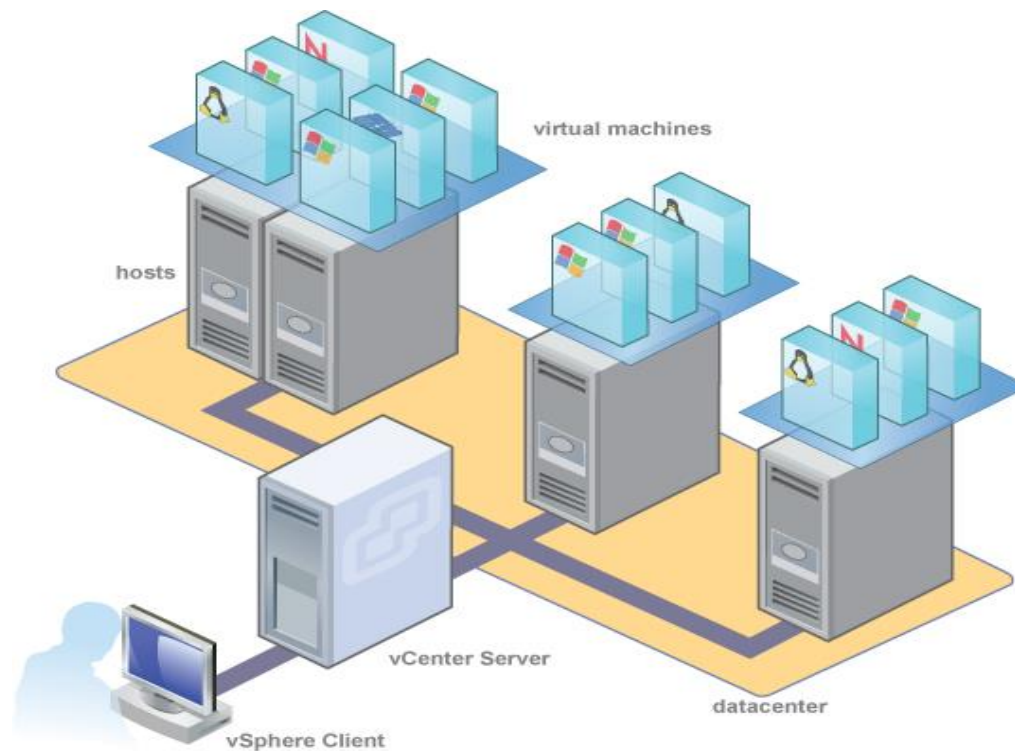
ESXi administration through vsphere Client



Vsphere client overview



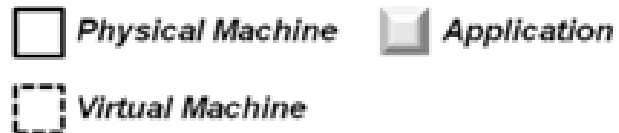
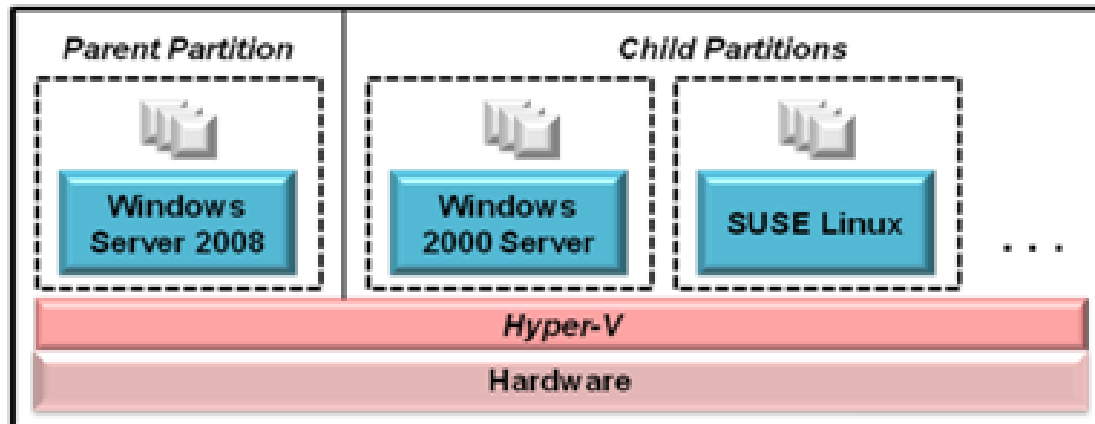
Administration of multiple ESXi_s through vCenter server



HYPER-v

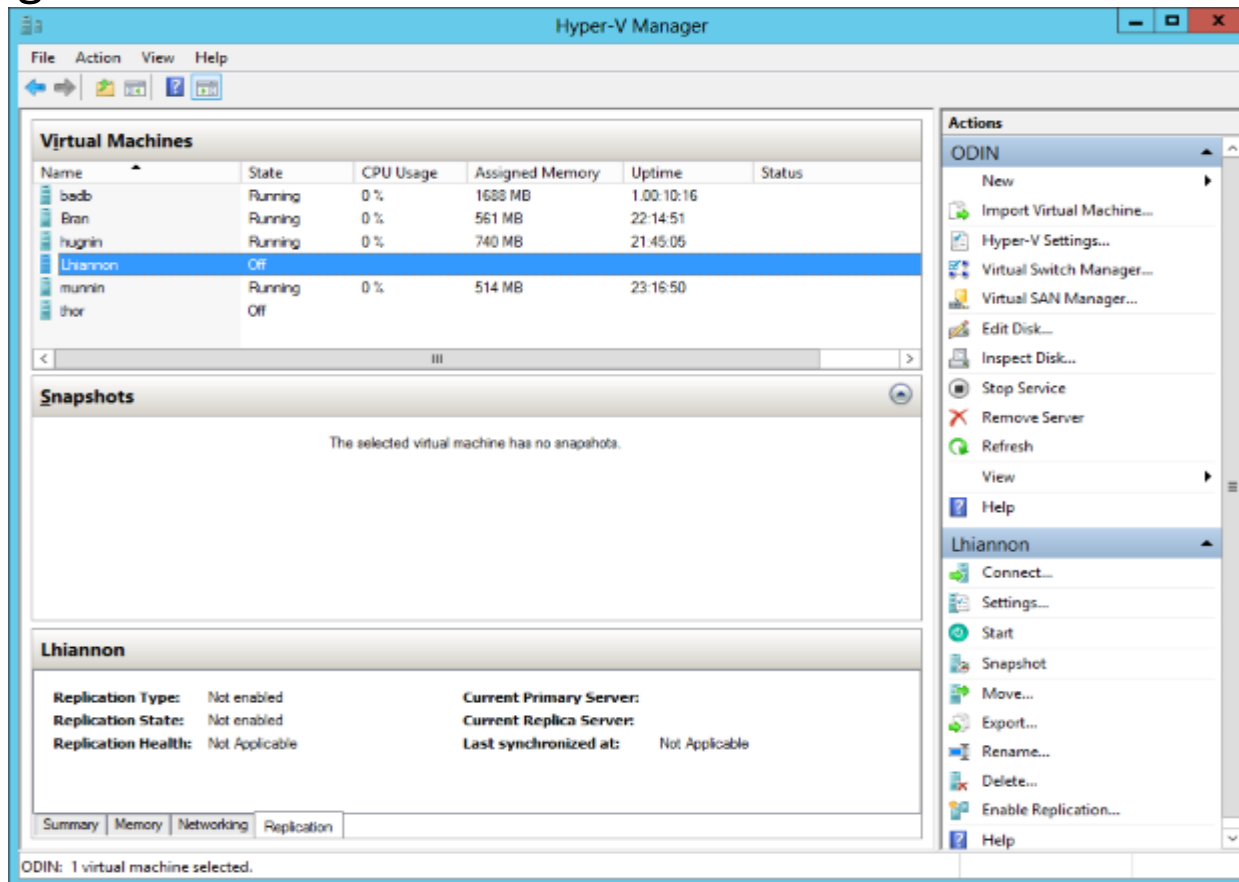
- Microsoft could not ignore the virtualization trend.
- Microsoft introduced Hyper-V as a virtualization platform in 2008, and it continued to release new Hyper-V versions with new Windows server versions.
- So far, there are a total of four versions, including
 - Windows Server 2012 R2,
 - Windows Server 2012,
 - Windows Server 2008 R2 and
 - Windows Server 2008.

HYPER-V



How HYPER-v Works?

- All Hyper-V management operations are available from Hyper-V Manager



Virtual Machine Creation

- Since Hyper-V role is also integrated into the Server Manager tool, one can find some useful information there by viewing the events and services related to Hyper-V and see recommended configurations and tasks.
- With Hyper-V Manager, one can create, change and delete VMs
- Modify virtual networking and perform additional dependent operations.

Virtual Machine Creation

- Since Hyper-V role is also integrated into the Server Manager tool, one can find some useful information there by viewing the events and recommended configurations.
- With Hyper-V Manager, one can create new virtual machines.
- Modify existing virtual machines.

